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MATLAB PROJECT

Title of the Project – Image Analysis using Matlab

#.Description of the project :→

#.About Image Analysis in matlab :→

→ Image analysis in MATLAB is a powerful area that leverages the capabilities of the **Image Processing Toolbox** to perform tasks such as image enhancement, filtering, segmentation, feature extraction, and more. It's widely used in fields like medical imaging, remote sensing, industrial inspection, and computer vision.

#.what are image Analysis used for ?

→ Image analysis is used in a wide range of real-world applications across industries. It's all about extracting meaningful information from images using computational techniques. Here's a breakdown of where and how it's used:

1) **1. Medical Imaging**

2) **Tumor detection** (MRI, CT, X-rays)
3) **Organ segmentation** (e.g., lungs, liver)
4) **Blood cell counting**
5) **Disease diagnosis** using AI-assisted image analysis

→Example: Detecting cancerous cells in histopathological images.

2. Automotive and Transportation

- **License plate recognition (LPR)**
- **Traffic monitoring**
- **Pedestrian and vehicle detection** (self-driving cars)

→ Example: Tesla's autopilot uses image analysis for lane detection and obstacle avoidance.

3. Satellite and Remote Sensing

- **Land cover classification**
- **Change detection (e.g., deforestation, urban growth)**
- **Disaster assessment (flood, fire, etc.)**

  → Example: Monitoring crop health using multispectral satellite images.

4. Industrial Automation / Manufacturing

- **Quality control** (detecting defects on assembly lines)
- **Pattern recognition and object counting**

→ Example: Checking if a product is sealed properly in packaging industries.

5. Forensics and Security

- **Face recognition and surveillance**
- **Biometric analysis (fingerprints, iris, etc.)**
- **Forgery detection**

→ Example: Analyzing CCTV footage for suspect tracking.

#. Our project is basically on the use of Image analysis using matlab .

We are going to detect moving vehicles of white colour on road .

→ Detecting a **white car on a road using MATLAB** is a cool and practical computer vision project! It typically involves color filtering, region segmentation, and possibly object detection using classical or AI-based approaches.

#.What are we going to use ?

→We are going to use Matlab Programming language .

#. Steps of Implementation

Step 1: **Access Video with VideoReader**

The VideoReader function constructs a multimedia reader object that can read video data from a multimedia file. See VideoReader for information on which formats are supported on your platform.

```
VideoReader with properties:

  General Properties:
        Name: 'traffic.mj2'
        Path: '/mathworks/devel/bat/filer/batfs2566-0/Bdoc24b.2855429/build/runnable/matlab/toolbox/images/imdata'
    Duration: 8
 CurrentTime: 0
  NumFrames: 120

  Video Properties:
       Width: 160
      Height: 120
   FrameRate: 15
 BitsPerPixel: 24
 VideoFormat: 'RGB24'
```

➔Code : trafficVid = VideoReader('traffic.mj2')

Step 2: **Explore Video with IMPLAY**

➔ Explore the video in implay.

Code : implay('traffic.mj2');

Step 3 : **Develop Your Algorithm**

➔When working with video data, it can be helpful to select a representative frame from the video and develop your algorithm on that frame. Then, this algorithm can be applied to the processing of all the frames in the video.

Code :

```
 darkCarValue = 50;
darkCar = im2gray(read(trafficVid,71));
noDarkCar = imextendedmax(darkCar, darkCarValue);
imshow(darkCar)
figure, imshow(noDarkCar)
```

→In the processed image, note how most of the dark-colored car objects are removed but many other extraneous objects remain, particularly the lane-markings.

```
sedisk = strel('disk',2);
```

```
noSmallStructures = imopen(noDarkCar, sedisk);
imshow(noSmallStructures)
```

Step 4: Apply the Algorithm to the Video

→ The car-tagging application processes the video one frame at a time in a loop.

#.code:

```
nframes = trafficVid.NumberOfFrames;
I = read(trafficVid, 1);
taggedCars = zeros([size(I,1) size(I,2) 3 nframes], class(I));

for k = 1 : nframes
   singleFrame = read(trafficVid, k);

   % Convert to grayscale to do morphological processing.
   I = rgb2gray(singleFrame);

   % Remove dark cars.
   noDarkCars = imextendedmax(I, darkCarValue);

   % Remove lane markings and other non-disk shaped structures.
   noSmallStructures = imopen(noDarkCars, sedisk);
```

```
    % Remove small structures.
    noSmallStructures = bwareaopen(noSmallStructures, 150);

    % Get the area and centroid of each remaining object in the
frame. The
    % object with the largest area is the light-colored car.  Create a
copy
    % of the original frame and tag the car by changing the centroid
pixel
    % value to red.
    taggedCars(:,:,:,k) = singleFrame;

    stats = regionprops(noSmallStructures,
{'Centroid','Area'});
    if ~isempty([stats.Area])
        areaArray = [stats.Area];
        [junk,idx] = max(areaArray);
        c = stats(idx).Centroid;
        c = floor(fliplr(c));
        width = 2;
        row = c(1)-width:c(1)+width;
        col = c(2)-width:c(2)+width;
```

```
        taggedCars(row,col,1,k) = 255;
        taggedCars(row,col,2,k) = 0;
        taggedCars(row,col,3,k) = 0;
    end
end
```

#.Step 5: Visualize Results

→ Get the frame rate of the original video and use it to see taggedCars in implay.

Code:

```
    frameRate = trafficVid.FrameRate;
implay(taggedCars,frameRate);
```

➔In this way we can finally complete the process of image analysis .

#. Output of our Project

➔White Car Detection in Video using MATLAB:-

This project demonstrates the detection of white-colored cars from a video stream using MATLAB. By applying color-based filtering techniques, specifically in

the HSV color space, the system isolates regions in the frame corresponding to white-colored objects. Morphological operations are then used to clean up the binary mask and eliminate noise. Finally, bounding boxes are drawn around the detected white cars, allowing for visual tracking in real-time. The approach can be extended to other color-based vehicle detections or combined with machine learning for improved accuracy in dynamic environments.

#. Advantage of this Project :-

1) **Real-Time Application**
2) **Simple and Efficient**
3) **Customizable**
4) **Low Computational Cost**
5) **Foundational for Advanced Projects**
6) **Educational Value**

#. Advantage of Image Analysis using matlab

1) **Comprehensive Toolbox Support**
2) **Rapid Prototyping:-**
   You can quickly develop and test algorithms

3)Wide Range of Functions

Offers hundreds of ready-to-use functions for tasks like:

→Filtering and enhancement

→Edge detection and segmentation

4)Interactive Apps

5) Integration with Machine Learning and Deep Learning

6) Automation & Batch Processing

#. Disadvantages of Image Analysis

:- As usual everything has some advantages and Disadvantages as well . Some of them are :-

1) Lighting Sensitivity:- Results can vary significantly with changes in lighting, shadows, or reflections

2) High Computational Cost (for advanced methods):-

3) Complexity in Real-World Scenarios:-

4) Requires Quality Data:- The effectiveness of image analysis depends heavily on the quality and resolution of input images. Poor-quality images may yield misleading results.

5) Limited Color-Based Approaches

6) Interpretation Challenges

7) Annotation Effort (for ML models)

Thanku So much !